• Enterprise AI · Infrastructure · Sustainability · June 2026

LLM Cascading: The *$500 Million* Wake-Up Call Every Enterprise Needed

> Why intelligent routing is the most impactful AI cost and sustainability decision your enterprise isn't making yet — and the research, incidents, and architecture that prove it.

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98% Max cost reduction proven by FrugalGPT (Stanford, 2023)	**80–90%** of all AI compute is now inference, not training	**21** peer-reviewed papers & reports cited in this document
$500M Burned in one month by a Fortune 500 with no usage limits (May 2026)	**945 TWh** Projected global data-centre electricity demand by 2030 — IEA	**16.6%** of production queries actually need a frontier model to achieve frontier-level accuracy

The Bill No One Budgeted For

The AI cost reckoning arrived faster than anyone expected. In the span of just a few months, a string of headline incidents illustrated what happens when enterprise AI deployments run without guardrails, routing intelligence, or developer education.

$500M

UNNAMED FORTUNE 500 · MAY 2026

A major corporation burned through approximately $500 million in Anthropic Claude credits in a single month after management failed to set usage limits when granting employees access to Claude accounts. The scale alone narrows it to only the very largest global corporations. *(Axios / Android Authority, May 2026)*

Full Year

UBER · APRIL 2026

Uber exhausted its entire 2026 AI coding tools budget in just four months. Claude Code adoption jumped from 32% to 84% of 5,000 engineers in weeks, with per-engineer spend reaching $500–$2,000/month. CTO Praveen Neppalli Naga confirmed the overrun and called it a "head-exploding moment." He personally spent $1,200 in a single two-hour demo session. *(The Information / Fortune, April 2026)*

$1.3M

OPENCLAW (3-PERSON TEAM) · MAY 2026

Running 100 parallel Codex agents, a three-person team burned through $1.3 million in OpenAI API tokens in 30 days — 60.3 billion tokens, 7.6 million requests. The bill was covered by OpenAI as an internal experiment. At standard rates it would have been $300K. *(X/Twitter disclosure, May 2026)*

$18,000

GOOGLE CLOUD USER · APRIL 2026

A developer woke up to an $18,000 Google Cloud bill after starting with a $7 budget — caused by an API key leaked in a public repository and abused overnight. A reminder that token spend is a live security surface, not just a cost concern. *(Google Cloud incident reports, April 2026)*

These are not edge cases. According to Zylo's 2026 SaaS Management Index, **78% of IT leaders experienced unexpected charges tied to AI and consumption**, and 60% lack full visibility into generative AI usage. The average enterprise AI budget has grown from $1.2M per year in 2024 to $7M in 2026. Some Fortune 500 companies report monthly AI inference bills in the tens of millions.

The concept of **"tokenmaxxing"** — where companies rank engineers on internal leaderboards by token consumption volume — directly accelerated Uber's overrun. Uber built exactly such a leaderboard. Meta uses similar dashboards. Critics argue high token consumption does not necessarily translate to better outcomes — but wasteful spending.

Meanwhile, Microsoft cancelled 200MW of US data-centre leases in February 2025, signalling concerns about AI infrastructure overcommitment, even as it maintained its $80B annual AI infrastructure pledge.

The Inference Energy Crisis That Deserves Far More *Attention*

Most conversations about AI's environmental footprint focus on training. The numbers are dramatic — training a large model can emit hundreds of tonnes of CO_2 equivalent. But training happens once. **Inference happens billions of times a day.** And inference is where the real operational footprint now lives.

80–90%
of all AI computing is now used for inference, not training — every query is a direct environmental act
MIT Technology Review, 2025

945 TWh
Projected global data-centre electricity demand by 2030 — more than Japan's entire consumption today
IEA Energy & AI Report, April 2025

79.7M t
Upper estimate of AI carbon footprint in CO_2eq by 2025 — equivalent to New York City's annual emissions
ScienceDirect / de Vries-Gao, 2025

The MIT Technology Review's landmark 2025 analysis concluded that **inference, not training, now represents the dominant and growing driver of AI energy usage**. Google estimated that 60% of AI energy in a typical deployment goes to inference; 40% to training. As model deployment scales, this ratio continues to shift further toward inference.

Goldman Sachs Research's August 2025 analysis added a troubling data point: roughly **60% of the increasing electricity demand from data centres will be met by burning fossil fuels**, adding an estimated 220 million tonnes of CO_2 to global emissions. Data centres already account for 4.4% of all US electricity consumption, and their carbon intensity is 48% higher than the US average.

WHY MODEL SIZE IS THE VARIABLE THAT MATTERS MOST

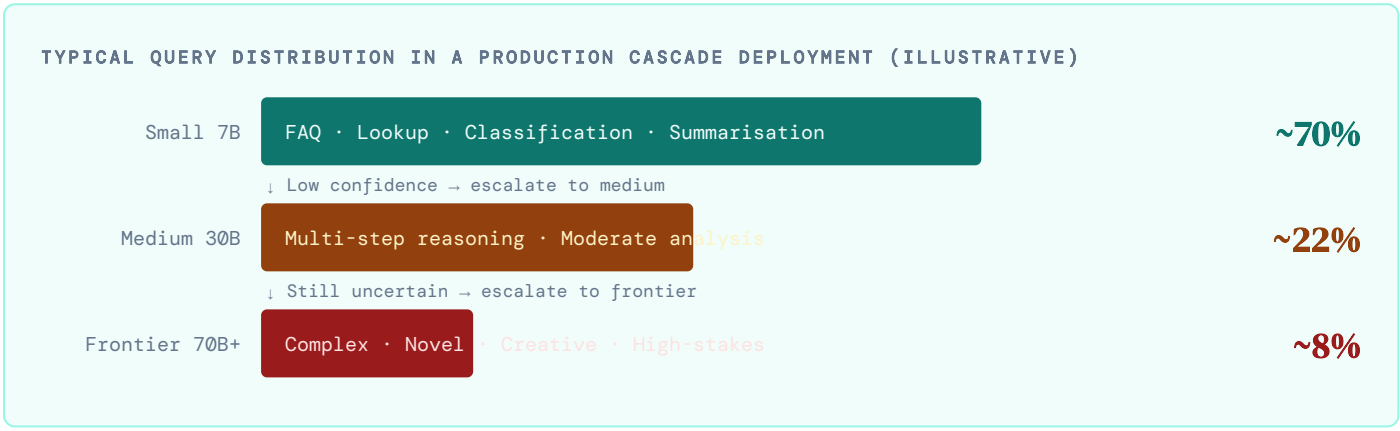
Larger models require more memory bandwidth per token, more active GPU compute, and longer decode times. Per-output-token energy in the decode phase accounts for over 96% of total inference energy per call. **A 70B-parameter model may consume 50–100× more energy per response than a 7B model answering the same simple question.** Every unnecessary frontier model call is not a hypothetical cost — it is a real, measurable, avoidable environmental event.

The IEA's April 2025 report *Energy and AI* projects data-centre electricity consumption will more than double by 2030. AI-focused data centres grew 50% in energy consumption in 2025 alone — growing even faster than total data-centre demand, which itself grew 17%. The efficiency gains in per-task energy are real, but total usage is growing faster than efficiency improves because more people are using AI for more energy-intensive tasks.

What LLM Cascading Actually Is

LLM cascading is a tiered routing strategy: each query is dispatched to the **smallest model capable of answering it with sufficient confidence**. Only queries that fall below a confidence threshold escalate to a more capable — and more expensive — tier. The frontier model is the last resort, not the default.

The architecture was formalised and validated by the landmark **FrugalGPT** paper from Stanford (Chen, Zaharia & Zou, 2023, arXiv:2305.05176), and extended by the open-source **RouteLLM** framework from UC Berkeley, Anyscale, and Canva (Ong et al., July 2024, arXiv:2406.18665).



The key empirical finding from FrugalGPT: **only 16.6% of queries needed to reach GPT-4** to achieve GPT-4-level accuracy across the tested benchmarks. The remaining 83.4% were handled confidently and correctly by cheaper models. RouteLLM achieved 95% of GPT-4 performance while requiring only 54% of GPT-4 calls.

The confidence gate: the hinge of the entire system

The entire value of a cascade depends on the quality of the confidence measurement. Four approaches, in order of increasing complexity: **Token probability scoring** — the model's output logits give a free confidence signal during inference. **Self-evaluation prompting** — after generating, prompt the model to rate its own confidence. **Classifier-based pre-routing** — a lightweight DistillBERT classifier trained on labelled query difficulty routes before any generative call is made. FrugalGPT demonstrated DistillBERT can accurately predict answer quality for models as large as GPT-4. **Speculative cascading** — a small model drafts; a verifier accepts or rejects without regenerating from scratch.

THE ESCALATION THRESHOLD IS YOUR MOST IMPORTANT HYPERPARAMETER

Run offline evals on a representative sample of production queries before setting thresholds. A threshold too aggressive escalates too frequently and defeats the purpose. Too conservative and quality degrades on the queries that matter. Starting point: 0.85 logit-based confidence for tier 1→2 escalation. Recalibrate quarterly — treat thresholds like model weights.

The Real Cost Math — And Why It Gets Worse

ROUTING STRATEGY	MODEL MIX	RELATIVE COST INDEX	VS. FRONTIER-ONLY
Frontier-only (no cascade)	100% large model	100	—
Conservative cascade	Small 50% / Medium 30% / Large 20%	28	↓ 72%
Aggressive cascade	Small 72% / Medium 20% / Large 8%	11	↓ 89%
FrugalGPT (empirical, published)	Learned adaptive routing	2–50	↓ 50–98%

FrugalGPT's published results across four benchmarks showed cost savings ranging from **50% to 98%** while matching or exceeding GPT-4 accuracy. On the HEADLINES financial news dataset, it reduced cost by 98% while *exceeding* GPT-4's accuracy — because certain query types are better served by cheaper, domain-calibrated models than a generalised frontier model. Only 16.6% of queries needed to reach GPT-4 to achieve GPT-4-level results across benchmarks.

RouteLLM (UC Berkeley / Anyscale, 2024) trained routers on human preference data from Chatbot Arena. Their best-performing router achieved **95% of GPT-4 performance while requiring only 54% of GPT-4 calls** on MMLU tasks with domain augmentation. The framework is open-source with four routing strategies: matrix factorisation, weighted Elo, BERT classifier, and causal LLM router.

THE COMING PRICING NORMALISATION RISK

Current enterprise API pricing is partly subsidised by venture capital and hyperscaler cross-subsidies. Anthropic spent $2.66B on AWS compute in the first nine months of 2025 against an estimated $2.55B in revenue. OpenAI spends $1.35 for every dollar earned — losses driven primarily by inference serving costs. **The FinOps Foundation's 2026 State of FinOps report explicitly flags pricing normalisation as "the next major enterprise AI budgeting shock,"** projecting 30–50% price increases within 12–24 months. Teams that build cascading into their stack now will be insulated. Teams that don't are carrying hidden risk.

A Gartner 2026 report found that while token pricing is expected to fall 90% by 2030, **cheaper tokens will not translate to cheaper enterprise AI** — because agentic models require far more tokens per task, and providers will not fully pass through cost reductions. The efficiency gain must come from architecture, not from waiting for price drops. This is the structural argument for cascading that holds regardless of what happens to per-token pricing.

> ***"The most dangerous assumption in enterprise AI budgeting is that cheaper models tomorrow will fix the runaway spend today."***

The Overlooked Problem: *Developer Culture*

Technology alone will not solve this. The more fundamental issue is that developers — and the organisations that incentivise them — have not internalised a basic principle: **not every query requires an LLM at all, let alone a frontier reasoning model.** The "tokenmaxxing" culture at Uber and Meta is a symptom: treating token spend as a proxy for productivity, and frontier model access as a status signal.

✓ USE AN LLM FOR THIS	✗ DO NOT USE AN LLM FOR THIS
Unstructured text understanding Summarising, classifying, or extracting meaning from free-form text where structure is ambiguous	**Exact lookups & rule evaluation** Database queries, key-value retrieval, deterministic rules — a SQL query costs $0 and is 100% reliable
Natural language generation Drafting, rewriting, translating where fluency and tone matter to the end user	**Arithmetic & calculation** LLMs are unreliable for precise math. Use a calculator, a formula, or a code function
Multi-step synthesis Tasks requiring reasoning across multiple documents, conflicting evidence, or ambiguous intent	**Pattern matching & parsing** Regex, keyword detection, structured data parsing — deterministic tools are faster and cheaper
Creative & generative tasks Generating options, variants, or novel outputs where human-like fluency is the deliverable	**Simple FAQ routing** If the answer is in a lookup table or knowledge base, retrieve it — don't regenerate it

And within the LLM tier: not every query needs a reasoning model

OpenAI's own published documentation is explicit: reasoning models like o3 and o3-pro are intended for *"complex problem solving, coding, scientific reasoning, and multi-step agentic workflows."* Their recommended pattern is to use a standard GPT-class model for execution tasks and a reasoning model only for planning and ambiguous multi-document synthesis.

THE CASCADE QUESTION EVERY DEVELOPER SHOULD ASK BEFORE EVERY CALL

Before calling any LLM: Is this a lookup, rule, or calculation? If yes — skip the model entirely.

Before calling a frontier LLM: Would a 7B model answer this with >85% reliability? If yes — route there first. It's 50–100× cheaper and usually equally accurate.

Before using a reasoning mode: Does this query require multi-step planning or synthesis across conflicting evidence? If not — use standard mode. Reasoning tokens are typically priced at 2–5× standard tokens and add significant latency with no quality benefit on simple tasks.

Before running an agent loop: Can this be accomplished in one or two well-scoped calls? Agentic loops compound token costs with every step and are particularly prone to runaway spend without hard loop limits.

A 2025 research paper (SMART: Self-Aware Agent for Tool Overuse Mitigation, arXiv:2502.11435) found that training models to be aware of when to use tools vs. parametric knowledge reduced tool use by 24% while **improving overall performance by 37%**. The same principle applies to reasoning modes: invoking heavy reasoning overhead on a simple task degrades efficiency without improving output quality.

Implementing Cascading in an Enterprise

Moving from single-model deployment to a governed cascade is a multi-layered change — technical, organisational, and cultural. Here is a practical seven-step roadmap with the evidence base behind each step.

1

Instrument before you optimise

Deploy an AI gateway or observability layer to capture: model used per call, token counts (input and output), response latency, and cost per call. You cannot optimise what you cannot see. A 2025 Deloitte report found that only 9% of organisations had a mature AI governance framework — and that visibility into consumption was the primary gap. Without instrumentation, you have no baseline to improve from.

2

Classify your query distribution

Sample 1,000–5,000 real production queries and classify them by complexity: simple (FAQ, lookup, classification), moderate (multi-step, moderate reasoning), and complex (ambiguous, novel, multi-document synthesis). This classification becomes the ground truth for your routing model. In most production deployments, 60–75% of queries fall into the "simple" bucket — the primary source of savings.

3

Define your tier architecture

Choose your model tiers based on your compliance posture, latency requirements, and data sovereignty rules. Enterprise frameworks like LiteLLM, Portkey, and Bifrost support on-premises models, Azure OpenAI, AWS Bedrock, and Anthropic APIs behind a single gateway. Enterprises with PII concerns can route sensitive queries to private/on-premises models while routing non-sensitive queries to cost-efficient public APIs — routing as compliance, not just cost control.

4

Build or adopt a router

RouteLLM (open-source, UC Berkeley) provides four pre-trained router strategies and an OpenAI-compatible server interface. For enterprise use cases with custom query distributions, train a domain-specific BERT classifier on your labelled query sample from step 2 — adding even 1,500 domain-specific samples produces significant performance improvements per RouteLLM's published results (arXiv:2406.18665).

5

Set budget controls at the gateway layer

Implement per-user, per-team, and per-application token budgets at the infrastructure layer — not the application layer. This is where the Uber and $500M incidents could have been caught. A gateway enforces these limits before the API call is made, not after the bill arrives. Set daily soft limits (notifications), weekly hard limits (throttle to smaller model), and monthly hard limits (suspend access). Monitor escalation rate as an operational metric.

6

Train developers — make it part of onboarding

Create an internal "AI usage guide" covering: when not to use an LLM, which tier to use for what task type, how to read your token dashboard, and how to set spending limits in IDE integrations. Uber's CTO spent $1,200 in two hours during a demo — not because the tool was broken, but because there was no internal culture of cost-consciousness. That culture is built by education, not by guardrails alone.

7

Track, iterate, and govern continuously

Review escalation rates, cost-per-query by team, and quality metrics monthly. A 2025 McKinsey report found that only 28% of organisations have C-suite or board-level oversight of AI governance — yet companies with formal frameworks report significantly greater AI value realisation. Treat your routing thresholds like model weights: they require periodic recalibration as your query distribution evolves.

The Tools Available Today

The cascading infrastructure stack is mature, well-researched, and largely open-source. Most enterprises are not using it. Here is the current landscape:

FrugalGPT

OPEN SOURCE · RESEARCH

Stanford's foundational cascading paper and implementation (Chen, Zaharia & Zou, 2023). Demonstrated 50–98% cost reduction across benchmarks. The academic baseline for all cascading work. Code and methodology publicly available.

arXiv:2305.05176

RouteLLM

OPEN SOURCE · UC BERKELEY

Open-source routing framework from UC Berkeley / Anyscale / Canva (July 2024). OpenAI-compatible server. Four router strategies: matrix factorisation, weighted Elo, BERT classifier, causal LLM. Routers and datasets published on HuggingFace.

arXiv:2406.18665 · github.com/lm-sys/RouteLLM

LiteLLM

OPEN SOURCE · PRODUCTION

OpenAI-compatible proxy supporting 100+ LLM providers. Built-in cost tracking, per-user/team budget controls, fallback routing, and rate limiting. Supports Azure OpenAI, AWS Bedrock, Anthropic, and self-hosted models behind one unified API.

github.com/BerriAI/litellm

Portkey AI Gateway

SAAS · ENTERPRISE

Enterprise AI gateway with intelligent routing, semantic caching (8% average cache hit rate with 99% cost savings on repeated prompts), fallback chains, and observability. Integrates FrugalGPT-style cascade patterns natively.

portkey.ai

Bifrost (Maxim AI)

OPEN SOURCE · GOVERNANCE

Open-source AI gateway with enterprise governance focus. Circuit-breaker routing, hierarchical budgets, compliance-aware routing (PII to private models), and audit logs aligned to NIST AI RMF and EU AI Act requirements.

getmaxim.ai/bifrost

GAR (Carbon-Aware)

RESEARCH · 2025

GAR (2025) adds carbon-aware routing via constrained optimisation — routes queries to lower-carbon inference regions while meeting SLA constraints. The first routing framework to explicitly optimise for environmental metrics alongside cost.

arXiv:2605.11603

Tradeoffs and failure modes to plan for

01 **Confidence miscalibration**

Small models can be confidently wrong on out-of-distribution queries. Logit-based scores are the most common signal but the least reliable for novel inputs. Run calibration evals on a held-out validation set before setting any production threshold.

02 **Tail latency on full escalation**

When a query escalates through all three tiers, total latency can be 3–5× worse than a direct frontier call. For latency-sensitive applications, set a hard escalation cap and use speculative/parallel calls for queries that would otherwise wait for sequential failures.

03 **Cascade collapse on adversarial inputs**

Jailbreak attempts and adversarial prompts cause small models to produce anomalously low-confidence outputs, flooding the frontier tier. Monitor escalation rate as an operational metric — a sudden spike is both a quality signal and a security alert.

04 **Distribution shift over time**

As your product evolves, the query distribution changes. A router calibrated on six-month-old data may be systematically miscalibrated. Schedule quarterly re-evaluations and treat routing thresholds like model weights — they need maintenance.

Reasoning Models Are Making Cascading *More* Important, Not Less

There is a widespread assumption circulating in engineering teams: that as models become more capable, the need for intelligent routing will diminish. The data says the opposite is happening. **Smarter models are, on average, dramatically more expensive per query — not less.**

TOKEN COST MULTIPLIER: STANDARD VS REASONING ERA (ORDER-OF-MAGNITUDE, ILLUSTRATIVE)

GPT-4 ERA · STANDARD MODEL

~800

typical visible output tokens per query
What you see = what you pay for.
Varies by task.

REASONING ERA · O3 / EXTENDED THINKING

10×+

typical total token multiplier (hidden + visible)
Visible tokens: similar to standard
+ hidden thinking tokens billed as output
Multiplier varies by task complexity.

> A request producing 500 visible output tokens may actually consume 5,000 tokens including the thinking budget — multiplying your effective output cost by **3–10× without warning**. *(BigGo Finance / Robonomics, April 2026)*

Research on Large Reasoning Models (arXiv:2506.23840, June 2025) documented a phenomenon called **"overthinking"**: even for trivially simple problems such as "9900 + 1 = ?", reasoning models produce outputs cluttered with extensive reflective thinking patterns exceeding thousands of tokens. The model thinks hard about easy questions because it does not know they are easy. **Identifying that a question is easy is exactly what a cascade router provides.**

Per-token prices are falling by 2–3× per generation. But output sizes are rising by 10× due to massive reasoning outputs. Overall cost per unit of operation actually grows while per-token prices drop. This is the core paradox of the reasoning era.

Agentic Loops: Where Costs Grow *Super-Linearly*

When a reasoning model runs inside an agentic loop — planning steps, calling tools, evaluating results, deciding next actions — token costs can grow super-linearly because each iteration often carries forward an expanding context window. Context history accumulates; each turn re-sends everything that came before. Research from Stevens Institute documents that a multi-step Reflexion loop can consume an order of magnitude more tokens than a single-pass call on the same task — not because the model is being wasteful, but because reasoning across accumulated context is genuinely what it is doing.

> ***"The strategic insight here is that one size does not fit all. A robust system must employ a routing pattern that classifies the complexity of an incoming query and routes it to the appropriate tier. Cheap and fast for simple queries. Deep and slow for complex ones."***
>
> — STEVENS INSTITUTE, HIDDEN ECONOMICS OF AI AGENTS, JANUARY 2026

OpenAI's own documentation confirms the tiered model approach: *"Most AI workflows will use a combination of both models — o-series for agentic planning and decision-making, GPT-series for task execution."* The cascade is not a workaround around reasoning models. It is the architecture that makes reasoning models **financially viable to deploy at scale.**

THE CASCADE-REASONING INTERACTION IN PRACTICE

Simple queries: Route to small standard model. No reasoning overhead. Sub-second, fraction of a cent. 99% cheaper than a reasoning call.

Moderate analysis: Route to medium standard model. Standard mode only — avoid o3-class for this. Minimal token overhead.

Novel multi-step reasoning: Frontier reasoning model justified. Extended thinking enabled. Accept token cost because the quality gain is real and measurable for this class of query.

Every generation, the small-model tier gets more capable for free. The frontier tier gets more powerful — and more expensive per task. **Cascading captures the value of both without paying the premium of either by default.**

Enterprise and Environment: *The Same Decision*

The sustainability case and the financial case are not in tension. They are the same case stated from different vantage points. If 70–80% of production queries are genuinely answerable by a small model, and each small-model call consumes 1/50th to 1/100th the energy of a frontier call, the carbon arithmetic is immediate: a well-implemented cascade can reduce the inference carbon footprint of an enterprise AI deployment by 50–70% without any degradation in output quality on the queries that matter.

For enterprise sustainability reporting: As ESG disclosure requirements tighten globally — particularly under EU CSRD and SEC climate disclosure rules — AI inference emissions are becoming a material reporting item. Building cascading into your AI stack is also building your emissions measurement and control posture. An AI gateway that tracks model-per-call data is simultaneously an audit trail for your carbon ledger. You get the sustainability reporting infrastructure for free as a byproduct of the cost governance infrastructure.

For enterprise financial governance: The incidents above — Uber, the unnamed $500M company, OpenClaw — all share the same root cause: unlimited access to frontier models with no routing intelligence and no consumption governance. Cascading solves the routing. Gateways solve the governance. Developer education solves the culture. None of these is optional in a mature AI deployment. All three must be in place for the system to work.

From Cloud FinOps to AI FinOps

Enterprises have been here before. In the early years of cloud computing, teams provisioned compute the way they had provisioned on-premise hardware: always on, billed to a cost centre that rarely looked at the invoice. Gartner estimated organisations wasted 30–35% of cloud spend through overprovisioning. The answer was FinOps — a practice that brought financial accountability to cloud infrastructure. **AI is following the same arc, compressed into a shorter timeframe.**

CLOUD FINOPS TRACKED	AI FINOPS MUST ALSO TRACK
Cost per compute-hour	**Cost per workflow** — not per API call, because one workflow = many calls
Resource utilisation rate	**Frontier model utilisation rate** — what % of calls actually needed the expensive tier?
Cost per team / department	**Cost per employee** — with alert thresholds, not just monthly reporting
Idle instance detection	**Escalation rate monitoring** — sudden spikes signal miscalibration or adversarial activity
Carbon / PUE reporting	**Carbon per inference** — model tier × energy intensity × grid carbon factor

None of these AI FinOps metrics exist by default in any provider's billing dashboard. They require instrumentation at the gateway layer — which is precisely where a cascade router already lives. **The team that builds its cascade also builds its AI FinOps observability for free.** The FinOps Foundation's 2026 State of FinOps Report (data.finops.org) identifies AI as the single top forward-looking priority, with token-based pricing and agentic billing introducing cost volatility that legacy frameworks cannot handle.

The War in the Middle East Just Made Every AI Query *More Expensive*

The US–Israel military strikes on Iran beginning February 28, 2026 sent an immediate shockwave through global energy markets — and the consequences flow directly into the cost of running AI at scale. **This is not an abstract geopolitical concern. It is a balance sheet event for every enterprise running frontier AI workloads.**

+25%
OIL PRICE SURGE

Global oil prices surged over 25% since the start of the conflict, with Brent crude breaching $80/barrel within days of the first strikes. *(Al Jazeera, March 2026)*

20%
GLOBAL OIL AT RISK

~20% of the world's daily oil supply — 20 million barrels — transits the Strait of Hormuz. Iran has threatened closure. Qatar declared force majeure on LNG exports. *(Statista / Reuters, March 2026)*

+267%
LOCAL ELECTRICITY COSTS

Electricity costs near large data-centre developments have risen up to 267% compared with 2020 levels — even before the conflict. *(Fortune / Morgan Stanley, Feb 2026)*

The direct link to AI inference cost

Data centres run on electricity. Electricity in most regions is priced with natural gas as the marginal fuel source. When oil and gas prices spike — as they have since February 28 — the operating cost of every GPU rack rises in lockstep. **Data centres and generative AI are forecast to account for 75% of incremental electricity load growth through 2030** (Morgan Stanley, 2026). That demand is already pushing electricity costs up in regions with high data-centre density — and an energy supply shock amplifies every existing pressure point simultaneously.

Europe faces a compounding risk: the cost of securing data-centre capacity in Europe's five largest markets — Frankfurt, London, Amsterdam, Paris, and Dublin — is set to rise 12% in 2026, before accounting for the geopolitical energy premium layered on top. **Every unnecessary frontier model call made during a period of energy price volatility is a cost that compounds twice** — once at the token price level, and again at the energy cost embedded in that token price.

WHY THIS MAKES CASCADING URGENT — NOT OPTIONAL

Geopolitical energy shocks are unpredictable. Per-token API prices — set by providers — do not adjust in real time to energy market conditions. But the cost pressure is real and will eventually be passed through. Enterprises that have already built intelligent routing into their stack are structurally insulated: they are running 70–80% of their queries on small, efficient models regardless of what happens to energy prices. Enterprises still routing everything to frontier models are fully exposed to every energy price shock that follows. The architecture decision you make today determines your exposure to every geopolitical event that affects energy prices tomorrow.

The smarter models become, the more important intelligent routing becomes.

Every generation of AI models produces more capable, more expensive frontier options — and simultaneously produces cheaper, more capable small models. GPT-5 mini at $0.25/M input tokens is more capable today than GPT-4 was in 2023, when it cost $30/M tokens. The small-model tier of your cascade keeps getting better for free. The frontier tier keeps getting more powerful — and when used thoughtlessly, more expensive per task.

The gap between "capable enough for this query" and "most powerful available" is growing in both directions simultaneously — more capability at the bottom, more token consumption at the top. Cascading is the architecture that captures the value of both without paying the premium of either by default.

The sustainability argument and the cost argument converge on the same architectural decision. Routing queries intelligently — smallest sufficient model first, frontier model last — happens to be both the frugal choice and the responsible one. In a world where AI inference workloads are growing faster than the grid is decarbonising, that alignment matters.

> ***"The most successful organisations will not be those with access to the most powerful models. They will be those that can determine when not to use them."***

The techniques are available now. FrugalGPT, RouteLLM, and a growing ecosystem of routing frameworks have made implementation tractable. The question for most teams is no longer *can we do this* — it is **why haven't we started yet.**

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Every Claim Is Backed by Data

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Writing about the systems, incentives, and infrastructure shaping enterprise AI. Drawing on experience across product development, emerging technologies, and large-scale AI initiatives, I explore what separates successful AI adoption from expensive experimentation. If this article sparked a thought, challenged an assumption, or inspired a question — let's connect.

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